

Booklet No.

Electrical Engineering Paper-I

Time Allowed : Three Hours

Maximum Marks : 200

Medium : English

Type of Paper : Conventional

QUESTION PAPER SPECIFIC INSTRUCTIONS

Please read each of the following instructions carefully before attempting questions :

1. There are **EIGHT** questions divided in two Sections, out of which **FIVE** are to be attempted.
2. Question no. 1 and 5 are compulsory. Out of the remaining questions, **THREE** are to be attempted choosing at least **ONE** question from each Sections.
3. The number of marks carried by a question/sub question is indicated against it.
4. Keep in mind the word limit indicated in the question if any.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
8. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
9. Neat sketches may be drawn, wherever required.
10. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.

Note - Candidates will be allowed to use Scientific (Non-programmable type) calculators.

SECTION - A

1. Answer **any five** of the following :

(8×5=40)

- (a) Use Ohms law and Kirchhoff's laws on the circuit given in fig. 1(a), find V_{in} , V_s and power provided by the dependent source.

(8)

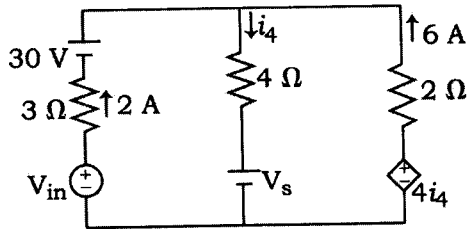


Fig. 1(a)

- (b) An alternating current varying sinusoidally, with a frequency of 50 Hz has an rms value of 20 A. Write down the equation for instantaneous value and find this value at (i) 0.025 s and (ii) 0.0125 s after passing through a positive maximum value. At what time, measured from a positive maximum value, will the instantaneous current be at 14.14 A ?

(8)

- (c) Find the

(8)

- (i) average value
- (ii) effective value
- (iii) form factor

of the saw tooth waveform shown in fig. 1(c)

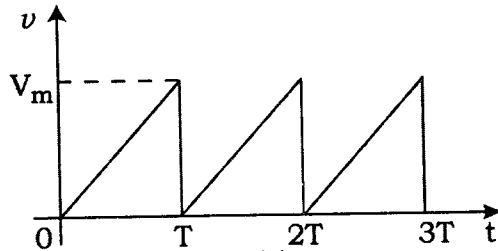


Fig. 1(c)

- (d) A long cylinder (fig. 1(d)) carries a charge density that is proportional to the distance from the axis $\rho = KS$, for some distance K . Find the electric field inside the cylinder.

(8)

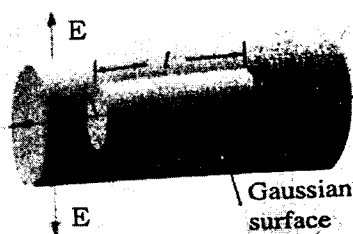


Fig. 1(d)

- (e) A mild steel ring having a cross-sectional area of 5 cm^2 and mean circumference of 40 cm has a coil of 200 turns wound uniformly around it. Calculate : (8)
- (i) reluctance of ring
- (ii) current required to produce a flux of $800 \mu\text{Wb}$ in the ring.
- Assume relative permeability of mild steel to be 380 at the flux density developed in the core.
- (f) A 6 pole, lap wound shunt motor has 500 conductors in the armature. The resistance of the armature path is 0.05Ω . The resistance of the shunt field is 25Ω . Find the speed of the motor when it takes 120 A from the dc main of 100 V supply. Flux per pole = 20 mWb . (8)
- (g) A 3 -phase, 6 pole induction motor delivers 3.73 kW at 950 r.p.m. What is the stator input, if the stator loss is 250 W ? (8)

2. (a) Determine the voltage at each node for the circuit shown in fig. 2(a). (15)

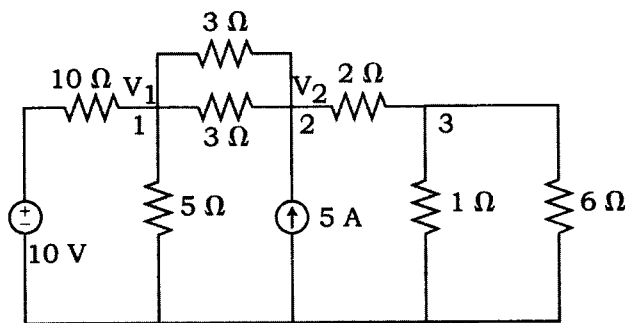


Fig. 2(a)

- (b) Use the nodal analysis to find the power dissipated in 6Ω resistor for the circuit shown in fig. 2(b). (15)

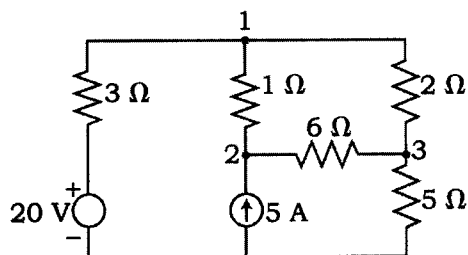


Fig. 2(b)

- (c) A hollow sphere is charged to $12 \mu\text{C}$ of electricity. Find the potential : (3)
- (i) at its surface (3)
- (ii) inside the sphere (4)
- (iii) at a distance of 0.3 meter from the surface. The radius of the sphere is 0.1 meter .

3. (a) What are the desired properties of magnetic materials used for the manufacture of electrical machines. (15)

Explain, why steel possessing higher silicon content is used in transformer while the sheet steel used in rotating machine has lower silicon content ?

- (b) A tap-changer transformer of a nominal ratio 11/33 kV supplies a load of 5000 kVA with lagging power factor of 0.8. The voltage at load end of the line is 30 kV, while $R=3\ \Omega$ and $X=2\ \Omega$ for each conductor. The mesh connected primary of the transformer has $R=0.17\ \Omega$ and $X=2.0\ \Omega$ per phase, while the star connected secondary has $R=0.5\ \Omega$ and $X=6\ \Omega$ per phase. Calculate the voltage at the secondary terminals and setting of the tap changer. (15)
- (c) A 1500 kVA, 6600 V, 3 phase star connected alternator with a resistance of $0.4\ \Omega$ and reactance of $6\ \Omega$ per phase, delivers full load current at power factor 0.8 lagging and normal rated voltage. Estimate the terminal voltage for the same excitation and load current at 0.8 power factor leading. (10)

4. (a) Calculate the effective inductance of the circuit shown in fig. 4(a) across AB. (15)

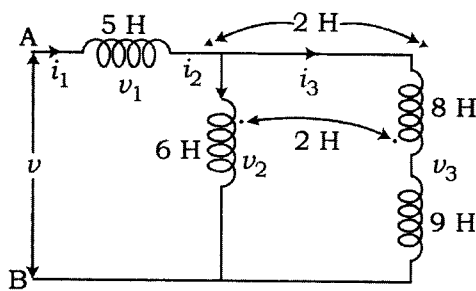


Fig. 4(a)

- (b) Explain the construction and principle of operation of BLDC motor. (15)
- (c) A 6 pole dc generator has 1000 shunt turns and 12 series turns per pole. For terminal voltage of 220 V on no load and 240 V. When supplying 300 A as shunt machine, the excitation current has to be increased from 2 to 5 A. Find the resistance of the diverter when the generator supplies this load as a compound machine. The series winding has a resistance of $0.005\ \Omega/\text{pole}$. (10)

SECTION – B

5. Attempt **any five** from seven. **(8x5=40)**
- (a) Explain the following phenomena in EHV system : **(4+4=8)**
- (i) Corona (ii) Radio Interference
- (b) Explain the first swing transient criteria and using the same prove equal area criterion. **(8)**
- (c) A synchronous motor improves the power factor of a load of 500 kW from 0.707 lagging to 0.95 lagging. Simultaneously the motor carries a load of 100 kW. **(3+3+2=8)**
- Find
- (i) the leading KVAR supplied by the motor,
- (ii) KVA rating of the motor and
- (iii) Power factor at which the motor operates.
- (d) Enlist the key achievements and challenges of the power sector in Maharashtra. **(8)**
- (e) (i) Define the terms : (I) Solid Angle (II) Candela (III) Luminous Efficiency and (IV) MSCP **(4)**
- (ii) An incandescent lamp rated 230 V takes 2.2 A and emits 8000 lumen. Calculate the efficiency in (I) MSCP per watt (II) lumen per watt. **(4)**
- (f) (i) State any four energy saving measures for DG set. **(4)**
- (ii) Connected load of a plant is 650 kW and Diversity Factor is 1.86. What is the desirable DG set rating with respect to 0.8 PF and set load factor of 70% ? **(4)**
- (g) A pump is used to fill a rectangular overhead tank measuring 5 m × 3.5 m with a height of 10 m. The inlet pipe to the tank is positioned at a height of 25 m above ground level. The pump draws water from an underground sump situated 4 m below the pump level and delivers it to a tank whose overflow line is positioned 8 m above the tank bottom. The motor driving the pump draws 7.5 kW of power. The operating efficiencies of the motor and the pump are 90% and 70% respectively. **(8)**
- Calculate the time taken by the pump to fill the tank up to the over flow level.

6. (a) A single-phase 50 Hz generator supplies an inductive load of 5000 kW at a PF of 0.707 lagging by means of an overhead transmission line 20 km long. The line resistance and inductance are 0.0195 ohm and 0.63 mH per km. The voltage at the receiving end is required to be kept constant at 10 kV. Find : (5+5+5=15)
- (i) the sending-end voltage and voltage regulation of the line.
 - (ii) the value of the capacitors to be placed in parallel with the load such that the regulation is reduced to 50% of that obtained in part (i)
 - (iii) Compare the transmission efficiency in part (i) and (ii).
- (b) Write a short note on **any three** of the following : (5+5+5=15)
- (i) Energy Policies
 - (ii) Smart Energy Meters
 - (iii) Fuel Cells and Super Capacitors
 - (iv) Energy systems for hybrid and electric vehicles
 - (v) Smart Grid System
- (c) Show that the L-G fault current on terminals of synchronous generators, the neutral of which is solidly connected is always greater than the three phase fault current on the terminal. Find the minimum value of neutral resistance in terms of sequence reactances of the machine to keep L-G fault current less than three phase fault current. (10)
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7. (a) (i) State the essential features of a good lighting scheme for interior of building. (5)
- (ii) An incandescent lamp hangs from the ceiling of a room. The illumination below the lamp vertically downwards is 80 lux. When the illumination is measured at a distance of 2 m from the vertical from the ceiling its value is 40 lux. Find the candle power of the lamp and its vertical distance from the floor. (10)
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- (b) (i) A centrifugal fan drawing 54 kW and operating at 1440 RPM is delivering air at 30,000 m³/hr. The head developed by the fan is 400 mmWC. If the speed is decreased by 200 RPM calculate the following :

(I) Air flow in m³/hr (1)

(II) Static Pressure in mmWC (2)

(III) Power drawn in kW (2)

- (ii) An energy audit in an industrial unit revealed a fan directly coupled with motor was operating at 37 Hz through VFD for 500 hours/month and supplying air through a 150 mm diameter duct. The fan is designed to deliver an air flow of 1300 m³/hr with a rated input power of 3 kW at 50 Hz. (10)

Calculate the air velocity and annual energy savings ignoring the motor losses.

- (c) A manufacturing unit operates a 180 kVA DG set rated at 0.8 lagging P.F. The prime mover is a diesel engine rated 240 BHP. The alternator has total losses (including exciter power) of 5.44 kW. Assume no derating for site conditions. The generator is required to supply a mixed industrial load at its full kVA rating. The plant manager wishes to improve system efficiency by operating at higher PF. The diesel engine operates at a brake thermal efficiency of 32% when loaded near its rated capacity. The calorific value of the diesel fuel is 10,500 kcal/kg and the specific gravity of the fuel is 0.85.

Calculate the following :

(i) Maximum PF that can be maintained at full kVA load without exceeding the engine capacity. (5)

(ii) Corresponding diesel fuel consumption (litres per hour) at this maximum power factor. (5)

8. (a) (i) State and deduce the relationship between two inertia constants (G and H) of the synchronous machine. (7)
- (ii) A 50 Hz, 4 pole turbo alternator rated 20 MVA, 11 kV has an inertia constant of $H=9$ kW-s/kVA. Find the acceleration if the input less than rotational losses is 26800 H.P. and power developed is 16 MW. (8)
- (b) (i) State the three types of UPS. Explain each types with the labelled block diagram. (9)
- (ii) State and explain the factors to be consider when selecting the right type of batteries for any project. (6)
- (c) (i) List five energy saving measures in compressed air system. (5)
- (ii) One of the machining centres has installed 2 No's of 270 cfm compressors for pneumatic operation and also for cleaning operation of components after machining. The compressor are operated at 7 kg/cm^2 and are on load for 80% of the time. The load power and the un-load power of each 270 cfm compressor is 40 kW and 15 kW respectively. The energy audit estimated that cleaning air requirement is 60% of the air generated. (5)
- Calculate the daily energy consumption for cleaning air alone assuming continuous operation of the compressor.

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